

CHENGZHI YAN

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Personal Homepage: <https://JacksonYan1.github.io>

RESEARCH INTERESTS

Artificial intelligence and machine-learning applications, with an emphasis on healthcare-related computing. Interested in data processing & engineering, time-series modelling, multi-source data integration, software system construction and AI-driven clinical auxiliary solutions.

EDUCATION

M.Sc. in Computer Science, University of Bristol, UK Sep. 2026 – Jun. 2027 (Expected)

Relevant Courses: Deep Learning, Sustainable Computing, Advanced Topics in Computer Science, Individual Research Project

B.Sc. in Computer Science, University of Bristol, UK Sep. 2022 – Jun. 2026

Relevant Courses: Artificial Intelligence, Software Engineering, Algorithms, Computer Systems, Computer Architecture

RESEARCH EXPERIENCE

Applied Data Science | Early Prediction of Sepsis, University of Bristol

Sep. 2025 – Jun. 2026 Supervisor: Prof. Saptarshi Sinha

- Developed machine-learning solutions for clinical predictive tasks using real-world patient datasets
- Performed full-cycle dataset processing: data cleaning, missing-value imputation, feature importance analysis and multi-dimensional data quality assessment
- Built reusable automated data processing pipelines; implemented hybrid time-series models including GRU-D + LightGBM and BRITS + LSTM
- Gained hands-on experience in complex dataset engineering and applying deep-learning techniques to domain-specific challenges

GitHub: <https://github.com/billu615/Applied-Data-Science-CW-Sepsis.git>

Software Engineering | Children's Mental Health Website, University of Bristol

Sep. 2023 – Sep. 2024 Supervisor: Prof. Sarah

- Collaborated to build a web-based resource platform for public-health-related content delivery
- Designed MongoDB data schemas and unified data entry standards; managed data models and multi-type content assets
- Implemented front-end modules with EJS and JavaScript; constructed backend services based on Node.js and Express.js
- Practised end-to-end web development, business-oriented data modelling and structured content management

GitHub: <https://github.com/spe-uob/2024-ThinkingSpace>

Team Project | Unity VR Accessibility Simulation, University of Bristol

Sep. 2025 – Jun. 2026 Supervisor: Prof. Sarah

- Developed an immersive VR simulation targeting accessibility-related research scenarios with Unity
- Managed heterogeneous assets including 3D scenes, character resources and physiological sensor signals
- Deployed LSTM-based time-series model for signal detection, integrated hardware

tactile feedback via haptic vest • Built practical engineering capabilities for multi-source heterogeneous-data integration and interactive system development

GitLab: https://gitlab.com/justice-246/bttf/-/tree/dev/game?ref_type=heads

Computer Systems | Conway's Game of Life Implemented in Go, University of Bristol

Sep. 2024 – Jan. 2025 Supervisor: Prof. Tilo

- Took part in project planning, literature review and formal technical report writing
- Implemented cellular-automaton simulation in Go, including unit testing and system-level performance optimisation
- Strengthened Go programming, performance tuning, literature survey and academic technical-writing competencies

GitHub: https://github.com/chw002/CSA_001.git

PATENTS

Granted Utility Model Patents

1. **Medical Radiation Protection Screen**, Patent No. ZL202421476455.6 | Granted 2024

Inventors: C. Yan (1st), X. Huang (2nd)

Contribution: Led mechanical-structure design for medical radiological shielding hardware.

[View Patent Certificate](#)

2. **Automatic Radiation Protection Screen**, Patent No. ZL202120693771.9 | Granted 2021

Inventors: Y. Yang (1st), C. Yan (2nd), J. He (3rd)

Contribution: Contributed to mechanical-structure and automatic-control design for radiology-protection equipment.

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3. **Electromagnetic Relay-Based Switching Control Circuit and Medical X-ray Protection Device**, Patent No. ZL202420775618.4 | Granted 2024

Inventors: D. Qin (1st), X. Huang (2nd), C. Yan (3rd)

Contribution: Participated in control-circuit logic design for medical X-ray protective devices.

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INTERNSHIP

Ruitai Imaging Technology (Shenzhen) Co., Ltd. | Full-stack Development Intern Digital Enterprise Inventory Management System (DingTalk Open-Platform Ecosystem)

- Conducted end-to-end requirement analysis, database schema design and full-stack application development
- Implemented procurement, warehousing and inventory-control modules leveraging DingTalk API with Node.js, MongoDB and JavaScript
- Replaced offline manual ledger workflows; boosted warehouse inbound-outbound efficiency by 40% and reduced manual-input errors, enabling unified visualised business-data management
- Accumulated practical experience building enterprise-grade data-centric systems and database optimisation for medical-related informatics research

SKILLS

Programming Languages • Proficient: Python, Go, C#, JavaScript • Skilled: C, Java • Familiar: EJS

AI & Machine Learning • Proficient: TensorFlow • Skilled: Scikit-learn, LightGBM • Familiar: Time-series models (GRU, LSTM, BRITS), dataset pre-processing, feature engineering, data-pipeline construction

Database, Systems & Development Tools • MongoDB, Express.js, Git/GitLab, Linux, Unity VR Development, Haptic hardware debugging • Data Visualization: Matplotlib, Seaborn

Research & Soft Skills • English-language literature retrieval, academic writing, experimental design, comparative experiment, dataset processing, group project coordination

Language Proficiency • Mandarin: Native • English: Able to read academic papers and draft independent research reports